

# **RIPHAH INTERNATIONAL UNIVERSITY**

**Gulberg Green Campus Islamabad**

**Faculty Of Computing**



## **SpiceSync**

**“Fast Food Management System”**

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**Course: Software Quality Engineering**

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# 1. INTRODUCTION

## 1.1 Project Overview

Spice-Sync is a Pakistan-focused fast food management system. This report presents test data generation and test case design for 3 core modules as part of Software Quality Engineering course requirements.

## 1.2 System Under Test

Spice-Sync supports 5 cities ( Islamabad, Karachi, Lahore, Multan, Abbottabad ) with features including user account management, order placement, cart management, and payment processing.

## 1.3 Testing Scope

| Module                      | Features | Test Cases |
|-----------------------------|----------|------------|
| M1: User Account Management | 10       | 100        |
| M2: Order & Cart Management | 10       | 100        |
| M3: Payment Processing      | 10       | 100        |
| Total                       | 30       | 300        |

Testing Techniques Used: Black Box Testing, Boundary Value Testing

# 2. SYSTEM REQUIREMENTS SUMMARY

## 2.1 Module M1: User Account Management (10 Features)

1. M1-F1: User Registration
2. M1-F2: User Login
3. M1-F3: OTP/Email Verification
4. M1-F4: Social Login
5. M1-F5: Forgot Password
6. M1-F6: Profile Management

7. M1-F7: Account Deletion
8. M1-F8: Notification Preferences
9. M1-F9: Session Management
10. M1-F10: Urdu Language Support

## **2.2 Module M2: Order & Cart Management (10 Features)**

1. M2-F1: Browse Menu & Item Discovery
2. M2-F2: Add to Cart & Quantity Management
3. M2-F3: Spice Level Customisation
4. M2-F4: Promo Codes & Discounts
5. M2-F5: Delivery Address & Location Selection
6. M2-F6: Group Order Feature
7. M2-F7: Reorder from History
8. M2-F8: Checkout & Order Confirmation
9. M2-F9: Order Cancellation
10. M2-F10: Nutritional Info & Allergen Display

## **2.3 Module M3: Payment Processing (10 Features)**

1. M3-F1: JazzCash Payment
2. M3-F2: Easypaisa Payment
3. M3-F3: Credit/Debit Card Payment
4. M3-F4: Cash on Delivery (COD)
5. M3-F5: Saved Payment Methods
6. M3-F6: Spice Points (Loyalty) Redemption
7. M3-F7: Refund Processing
8. M3-F8: Payment Receipt Generation
9. M3-F9: Transaction History
10. M3-F10: Payment Security & Fraud Prevention

### 3. TEST DATA GENERATION

Test data created with Valid, Invalid, and Boundary values for each field.

#### 3.1 Test Data for User Account Management

| Field    | Valid Data                           | Invalid Data                       | Boundary Values                |
|----------|--------------------------------------|------------------------------------|--------------------------------|
| Email    | user@gmail.com, ali.khan@outlook.com | user@, @gmail, plain, null, spaces | 64-255 chars                   |
| Password | Spice@2025, P@ss#99                  | 12345, pass, abc, empty            | 8 chars (min), 128 chars (max) |
| Phone    | +92-300-1234567, +92-321-9876543     | 03001234, null, 7 digits           | +92XXXXXXXXXX (13 chars total) |
| Username | ali_99, fatima.noor                  | ab (too short), ali!!, spaces      | 3-30 chars                     |
| OTP Code | 123456 (valid 6-digit)               | 12345, abcdef, empty, 1234567      | Exactly 6 digits               |

#### 3.2 Test Data for Order & Cart Management

| Field       | Valid Data                            | Invalid Data                         | Boundary Values                   |
|-------------|---------------------------------------|--------------------------------------|-----------------------------------|
| Quantity    | 1, 5, 10, 20                          | 0, -1, 21, 999, 'five' (text)        | Exactly 20 (max), Exactly 1 (min) |
| Spice Level | Mild, Medium, Hot, Peshawari, Punjabi | Invalid string, null, numeric 7      | All 6 predefined levels           |
| Promo Code  | EID2025 (10%), RAMADAN20 (20%)        | EXPIREDCODE, XXXXX, empty, used code | At discount cap                   |
| City        | Islamabad, Karachi, Lahore, Multan    | Quetta, 'Mars', null, numeric        | All 5 supported cities            |

#### 3.3 Test Data for Payment Processing

| Field           | Valid Data                   | Invalid Data                               | Boundary Values                |
|-----------------|------------------------------|--------------------------------------------|--------------------------------|
| JazzCash MPIN   | 03001234567, 4-digit: 1234   | Wrong PIN, non-JazzCash number             | PKR 10,000 max per transaction |
| Easypaisa OTP   | 03211234567, 6-digit: 654321 | Non-Easypaisa number, Expired OTP (>3 min) | PKR 25,000 max per transaction |
| Card Number CVV | 4111111111111111, 123, 1234  | Luhn-fail: 1112, 00, letters, empty        | 16-digit card, 3-4 digits CVV  |
| COD Amount      | PKR 100 - 5000               | PKR 5001+                                  | Exactly PKR 5000               |

## 4. TEST CASE DESIGN

### 4.1 Test Case ID Convention

- TC-[Module ID]-[Feature ID]-[Sequence]
- Example: TC-M1F3-05 (Module 1, Feature 3, Test Case 05)

### 4.2 Test Type Colour Code

- Functional (Blue)
- Negative (Red)- UI/UX (Purple)
- Security (Brown)- Performance (Green)

### 4.3 Test Case Distribution

| Test Type   | Count | Percentage |
|-------------|-------|------------|
| Functional  | 128   | 42.7%      |
| Negative    | 90    | 30.0%      |
| UI/UX       | 41    | 13.7%      |
| Security    | 23    | 7.6%       |
| Performance | 18    | 6.0%       |
| TOTAL       | 300   | 100%       |

### 4.4 Sample Test Case Format

| Field       | Description                                            |
|-------------|--------------------------------------------------------|
| TC-ID       | Unique identifier (e.g., TC-M1F1-04)                   |
| Type        | Functional / Negative / UI/UX / Security / Performance |
| Description | What is being tested                                   |
| Test Data   | Input values to be used                                |
| Steps       | Step-by-step execution procedure                       |
| Expected    | Expected result                                        |
| Status      | Pass / Fail / Blocked / Not Executed                   |

## 4.5 Sample Test Cases

### TC-M1F1-04: Register with invalid email format

- Type: Negative
- Test Data: Email: user@, Password: Spice@2025
- Expected: "Please enter a valid email address"

### TC-M1F1-05: Register with weak password

- Type: Negative
- Test Data: Password: 12345
- Expected: "Password must be 8+ chars with upper, lower, digit & symbol"

### TC-M2F2-07: Add item with quantity exceeding maximum

- Type: Boundary
- Test Data: Quantity: 21
- Expected: "Maximum 20 units per item allowed"

### TC-M3F1-05: JazzCash payment with insufficient balance

- Type: Negative
- Test Data: Order PKR 1200, Balance PKR 500
- Expected: "Insufficient JazzCash balance"

## 5. VERIFICATION & VALIDATION (V&V)

### 5.1 Verification Activities (Static Testing)

| Activity            | Artifact       | Method      | Outcome           |
|---------------------|----------------|-------------|-------------------|
| Requirements Review | 30 Features    | Peer Review | Clear & testable  |
| Test Case Review    | 300 Test Cases | Walkthrough | 12 cases improved |
| Test Data Review    | Test Tables    | Inspection  | Complete coverage |

### 5.2 Validation Activities (Dynamic Testing - Planned)

| Test Type  | Test Cases | Status           |
|------------|------------|------------------|
| Functional | 128        | Ready to execute |



| Test Type   | Test Cases | Status           |
|-------------|------------|------------------|
| Negative    | 90         | Ready to execute |
| UI/UX       | 41         | Ready to execute |
| Security    | 23         | Ready to execute |
| Performance | 18         | Ready to execute |

## 6. DEFECTS LOG

### 6.1 Defects Summary

| Defect ID | Module | Severity | Description                                |
|-----------|--------|----------|--------------------------------------------|
| D-001     | M1-F3  | Medium   | OTP resend timer not resetting correctly   |
| D-002     | M1-F6  | Low      | Profile photo upload WebP display issue    |
| D-003     | M1-F10 | Low      | Partial Urdu translation coverage          |
| D-004     | M2-F3  | High     | Spice level price delta not applied        |
| D-005     | M2-F4  | High     | Promo code can be used multiple times      |
| D-006     | M2-F6  | High     | Large group order timeout handling         |
| D-007     | M3-F1  | Critical | Payment after MPIN blocked                 |
| D-008     | M3-F2  | Medium   | OTP expiry message wrong attempts          |
| D-009     | M3-F5  | Low      | Delete confirmation wrong card digits      |
| D-010     | M3-F10 | Critical | Payment amount can be modified client-side |

### 6.2 Defects by Severity

| Severity | Count |
|----------|-------|
| Critical | 2     |
| High     | 3     |

| Severity | Count |
|----------|-------|
| Medium   | 2     |
| Low      | 3     |
| TOTAL    | 10    |

## 7. ROOT CAUSE ANALYSIS (RCA)

### 7.1 RCA for Defects

| Defect ID | Root Cause                           | Category        |
|-----------|--------------------------------------|-----------------|
| D-001     | Race condition in timer logic        | Code Logic      |
| D-004     | Missing price delta calculation      | Code Omission   |
| D-005     | Missing unique usage tracking        | Requirement Gap |
| D-006     | No timeout handling for large groups | Design Flaw     |
| D-007     | Missing state check after failure    | Code Logic      |
| D-010     | Missing server-side validation       | Security Gap    |

### 7.2 RCA Distribution

| Category        | Percentage |
|-----------------|------------|
| Code Logic      | 35%        |
| Security Gap    | 30%        |
| Requirement Gap | 20%        |
| Design Flaw     | 15%        |

## 8. IMPROVEMENTS & PREVENTION TECHNIQUES

### 8.1 Short-term Improvements (Immediate Fixes)

| Defect | Proposed Fix                                     | Priority |
|--------|--------------------------------------------------|----------|
| D-005  | Add database table to track promo usage per user | High     |
| D-007  | Add account lock flag check before payment       | Critical |
| D-010  | Move all amount validation to server-side        | Critical |

## 8.2 Medium-term Improvements (Next Sprint)

1. Add Idempotency Keys for all payment transactions
2. Implement Rate Limiting on OTP resend endpoint
3. Add Distributed Locking for group orders >50 participants
4. Improve Urdu translation to 100% coverage

## 8.3 Long-term Prevention Techniques

| Technique             | Target              |
|-----------------------|---------------------|
| TDD                   | All new features    |
| Code Review Checklist | All pull requests   |
| Static Analysis       | Daily builds        |
| Security Scanning     | Weekly              |
| Load Testing          | Before each release |

## 8.4 QA Process Improvement

**Current:** Requirements → Coding → Testing → Release

**Proposed:** Requirements → [Review] → Coding → [Code Review] → Testing → [Security Scan] → QA Sign-off → [Performance] → Release

# 9. QUALITY METRICS

## 9.1 Test Coverage

$$\begin{aligned} \text{Test Coverage} &= (\text{Test Cases Designed} / \text{Test Cases Planned}) \times 100 \\ &= (300 / 300) \times 100 = 100\% \end{aligned}$$

## Table:

| Test Type   | Designed | Planned | Coverage |
|-------------|----------|---------|----------|
| Functional  | 128      | 128     | 100%     |
| Negative    | 90       | 90      | 100%     |
| UI/UX       | 41       | 41      | 100%     |
| Security    | 23       | 23      | 100%     |
| Performance | 18       | 18      | 100%     |

### 9.2 Defect Density

Defect Density = Number of Defects / Size (Features)  
= 10 defects / 30 features  
= 0.33 defects per feature

Industry Benchmark: 0.5-1 defects per feature = GOOD

### 9.3 Requirement Coverage

Requirement Coverage = (Features with Test Cases / Total Features) x 100  
= (30 / 30) x 100 = 100%

### 9.4 Metrics Dashboard

| Metric                 | Value | Status    |
|------------------------|-------|-----------|
| Test Coverage          | 100%  | Excellent |
| Requirement Coverage   | 100%  | Excellent |
| Defect Density         | 0.33  | Good      |
| Test Data Completeness | 100%  | Excellent |

## 10. CONCLUSION

### 10.1 Project Summary

| Aspect                  | Result                            |
|-------------------------|-----------------------------------|
| Requirements Documented | 30 Features across 3 modules      |
| Test Cases Designed     | 300 (10 per feature)              |
| Test Data Generated     | Complete (valid/invalid/boundary) |
| Defects Identified      | 10 (during test design phase)     |
| Test Coverage           | 100%                              |
| Defect Density          | 0.33 per feature                  |